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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

EE2301 Electronic Devices and Circuits Lab

Project

BY

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1 Objective

Microphone Amplifier with Class AB Driver

Objective

Design and test an audio amplifier chain.

Tasks

- Build a low-noise microphone pre-amplifier.
- Couple it to a Class AB power driver stage.
- Drive a small speaker and test sound quality, distortion, and efficiency.

Expected Outcome

A functional audio amplification system with bias stability and acceptable distortion.

Characterization Requirements

You must properly characterize your design with meaningful specifications such as:

- Total Harmonic Distortion (THD)
- Noise
- Power Supply Rejection Ratio (PSRR)
- Quiescent current (IQ)
- Gain and bandwidth
- Output swing and power

2 Components Required

1. Microphone Pre-Amplifier Stage

Active Components

- Electret condenser microphone (ECM)
- Q1 – BC547 NPN transistor
- Q2 – BC547 NPN transistor

Resistors

- R1 – 1.2 k Ω
- R2 – 15 k Ω
- R3 – 15 k Ω
- R4 – 3.9 k Ω
- R5 – 390 Ω
- R6 – 100 k Ω
- 50 k Ω - SET GAIN
- 2.2 k Ω microphone bias resistor

Capacitors

- C1 – 10 μF
- C2 – 10 μF
- C3 – 10 μF

Power Supply

- +12 V DC supply
- Ground

2. Class-AB Output Stage

Active Components

- BC547 / BC548 NPN transistor (driver stage)
- TIP31C NPN power transistor
- TIP32C PNP power transistor

Diodes

- Two biasing diodes – 1N4007

Resistors

- Base resistor for driver transistor
- Base resistors for TIP31C / TIP32C
- Emitter resistors as required for output stabilization

Capacitors

- 1000 μF (output coupling capacitor)
- Supply decoupling capacitors for the 12 V rail

Load

- Loudspeaker ($8\ \Omega$ or $16\ \Omega$)
- Optional series resistor for speaker protection

Power Supply

- +12 V DC supply
- Ground

3. Test and Measurement Equipment

- 12 V DC power supply
- Function generator
- Oscilloscope with FFT capability
- Digital multimeter
- Breadboard and connecting wires

3 Theory

1. Microphone Pre-Amplifier (Low-Noise CE Stage)

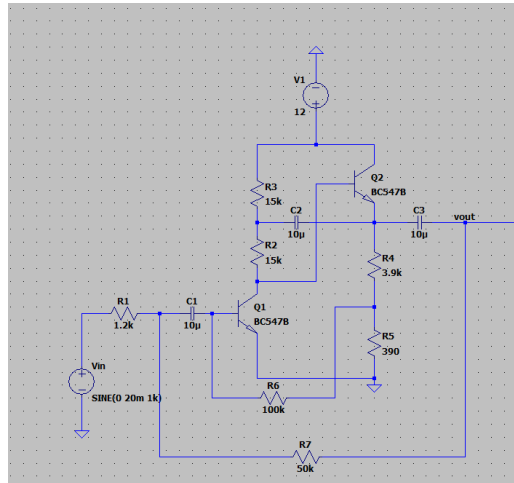


Figure 1: Circuit of Pre-Amplifier

- The electret condenser microphone (ECM) outputs a low level AC signal (5 to 20 mV) and requires a DC bias to power its internal FET.
- A common-emitter (CE) transistor stage provides high voltage gain and performs **impedance bridging**, ensuring minimal loading on the microphone.
- The input coupling capacitor blocks DC and forms a high pass network with the input impedance, improving low frequency response and PSRR.
- Base bias resistors establish the **quiescent operating point** of the transistor for linear amplification.
- The emitter resistor adds **thermal stability** and negative feedback; its bypass capacitor restores AC gain.
- The collector resistor forms a high-impedance load, maximizing gain.
- **Bootstrapping** techniques increase the effective input impedance and reduce noise by forcing nearly equal AC voltages across bias resistors.
- The pre-amplifier boosts the microphone signal to a level suitable for driving the Class AB power stage.

2. Class-AB Output Driver Stage (Push Pull Power Amplifier)

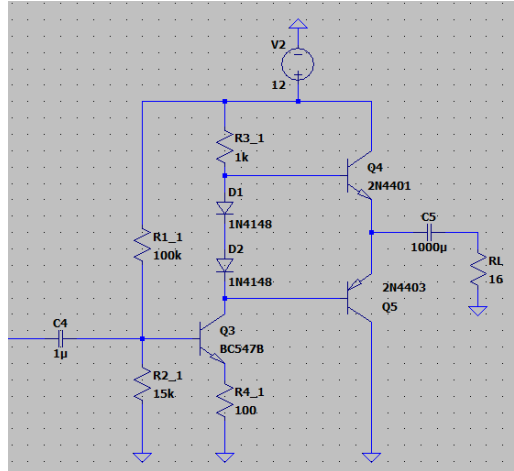


Figure 2: Circuit of Class-AB

- A complementary push pull pair (TIP31C NPN, TIP32C PNP) forms a **Class-AB** output stage capable of supplying high current to an 8 Ω loudspeaker.
- Two forward-biased diodes provide **Class-AB biasing**, maintaining a small quiescent current and eliminating **crossover distortion**.
- The NPN device amplifies positive half-cycles while the PNP handles negative half-cycles, enabling symmetrical current delivery.
- Emitter resistors- improve **thermal stability**, prevent current hogging, and linearize large-signal operation.
- The driver transistor feeds the bases of the power devices, ensuring adequate voltage swing and current gain.
- A large output coupling capacitor blocks DC and passes AC, forming a high-pass network with the speaker load.
- Supply decoupling capacitors enhance **PSRR** by suppressing ripple and preventing supply noise from entering the audio path.
- The Class-AB stage provides high efficiency, low THD, adequate bandwidth, and strong output drive for audio amplification.

4 Circuit

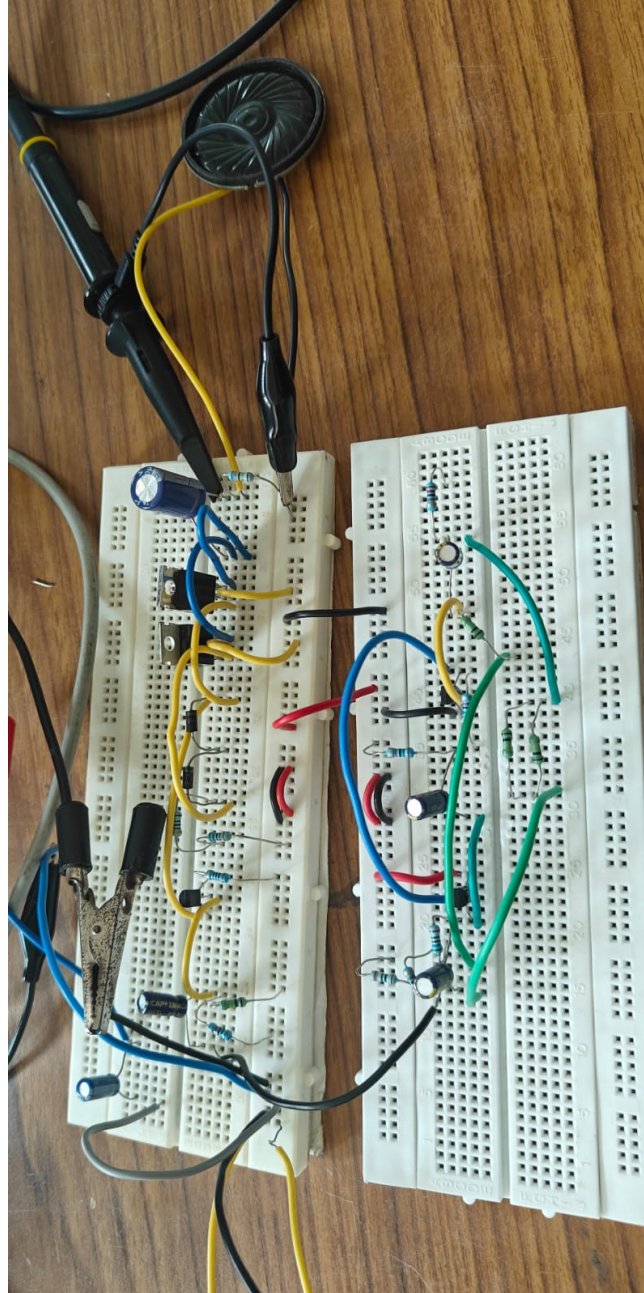


Figure 3: Circuit of CS

5 Procedure

1. Pre-Amplifier Testing

1. Connect the ECM with a $2.2\text{ k}\Omega$ bias resistor and couple it through the input capacitor.
2. Build the pre-amplifier using BC547 as per the circuit above.
3. Power the circuit from 12 V and check that the transistor is correctly biased by measuring V_B , V_C , and V_E .
4. Apply a small test signal (20 to 30 mV, 1 kHz) instead of the mic and observe the amplified output..

2. Class-AB Output Stage

1. Assemble the push pull stage using TIP31 and TIP32 with two diodes for Class-AB bias.
2. Power the circuit and measure the quiescent current to confirm proper AB operation.
3. Feed a 1 kHz sine wave (50 to 100 mV) to the input and connect an $8\text{ }\Omega$ load.
4. Observe positive and negative half cycles and ensure there is no crossover distortion at zero crossing.

3. Full Amplifier Test

1. Connect the pre-amp output to the Class-AB stage using a coupling capacitor.
2. Power both stages with the same 12 V supply and add decoupling capacitors.
3. Apply a mic or small test signal and check the loudspeaker output for clean audio.

4. Performance Measurements

4.1 Gain and Bandwidth

1. Give a fixed input (20 mV) and sweep frequency from 20 Hz to 1 MHz.
2. Record the output amplitude and find the -3 dB cutoff points.

4.2 THD

1. Apply a 1 kHz sine wave and keep output in the linear region.
2. Use oscilloscope FFT to note harmonics and calculate THD.

4.3 Noise

1. Short the input and measure the output noise level in AC mode.

4.4 PSRR

1. Inject a small ripple (around 100 mV) into the 12 V supply.
2. Measure ripple at supply and output and compute PSRR.

4.5 Output Swing and Power

1. Increase input until clipping starts and note the maximum clean output.
2. Calculate output power using $P = V_{\text{rms}}^2 / R_L$.

6 Observation

6.1 Quiescent Current (IQ)

With a 12 V supply and no input signal, the quiescent current of the Class AB stage was measured using a multimeter in series with the supply line. The measured value is:

$$I_Q = 98.7 \text{ mA}$$



Figure 4: Measurement of quiescent current using a digital multimeter.

This current corresponds to the standing Class AB bias set by the diode network.

6.2 Gain and Bandwidth

A sinusoidal input of 20 mV_{pp} was applied and the output amplitude was measured across frequency. The midband output level was around 3.0 V_{pp} , giving a voltage gain:

$$A_v = \frac{3.0}{0.020} \approx 150$$

The -3 dB level corresponds to:

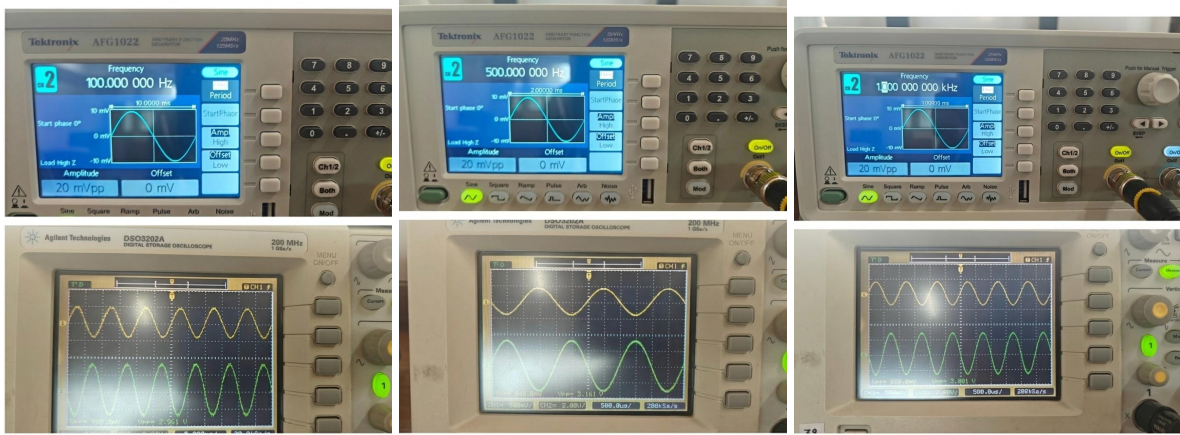
$$V_{3dB} = \frac{3.0}{\sqrt{2}} \approx 2.12 \text{ V}_{pp}$$

From the data, the output falls below 2.12 V_{pp} at approximately 550 600 kHz, giving a bandwidth of:

$$\text{Bandwidth} \approx 20 \text{ Hz to } 600 \text{ kHz}$$

Table 1: Frequency response and voltage gain

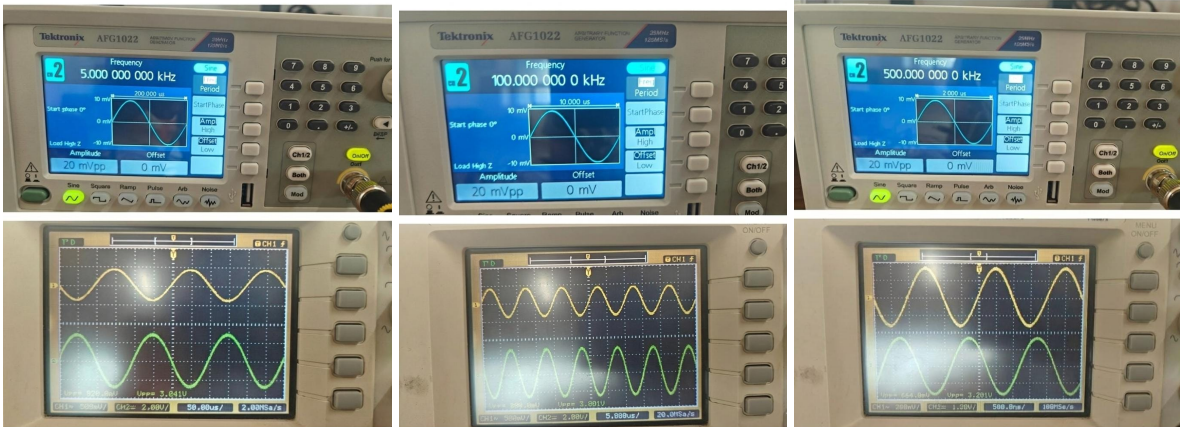
Frequency (Hz)	V _{out,pp} (V)	Gain $A_v = \frac{V_o}{0.020}$
20	2.201	110.05
50	2.801	140.05
100	2.961	148.05
500	3.161	158.05
1k	3.001	150.05
2k	3.001	150.05
5k	3.041	152.05
10k	3.082	154.10
50k	3.441	172.05
100k	3.801	190.05
200k	4.432	221.60
300k	4.081	204.05
400k	3.681	184.05
500k	3.201	160.05
550k	2.401	120.05
600k	1.601	80.05
630k	1.481	74.05
700k	1.801	90.05
800k	1.761	88.05
900k	1.561	78.05
1M	1.321	66.05
2M	0.440	22.00



(a) 100 Hz

(b) 500 Hz

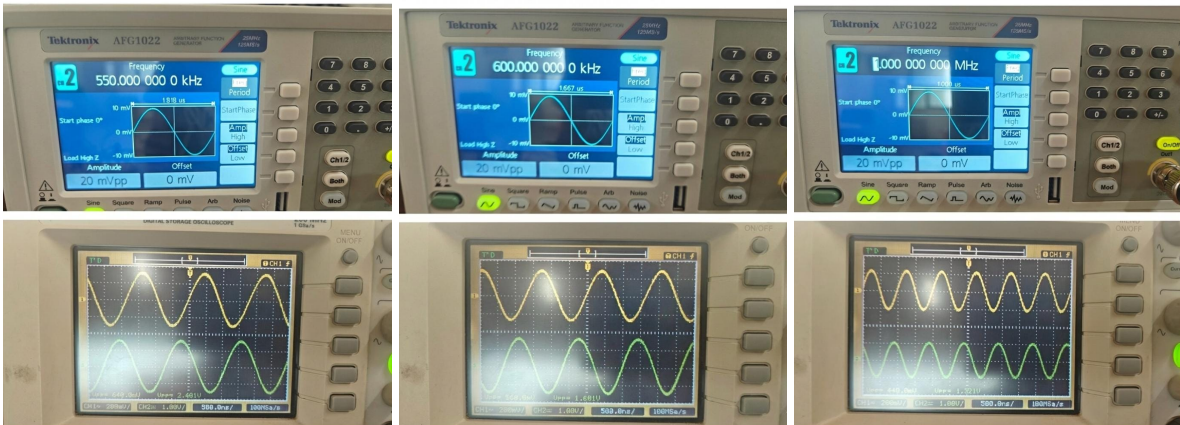
(c) 1 kHz



(d) 5 kHz

(e) 100 kHz

(f) 500 kHz



(g) 550 kHz

(h) 600 kHz

(i) 1 MHz

Figure 5: Output waveforms at selected frequencies for gain and bandwidth analysis.

6.3 Output Swing and Power

The output swing was measured at 1 kHz by increasing the input amplitude until visible clipping occurred. The corresponding output voltages and calculated RMS values are listed in Table 2. The maximum clean output swing was approximately 6.0 V_{pp}.

$$V_{rms} = \frac{6.0}{2\sqrt{2}} = 2.12 \text{ V}_{rms}$$

For an 8 Ω load, the maximum undistorted output power is:

$$P_{max} = \frac{(2.12)^2}{8} \approx 0.56 \text{ W}$$

Table 2: Input vs Output Swing at 1 kHz

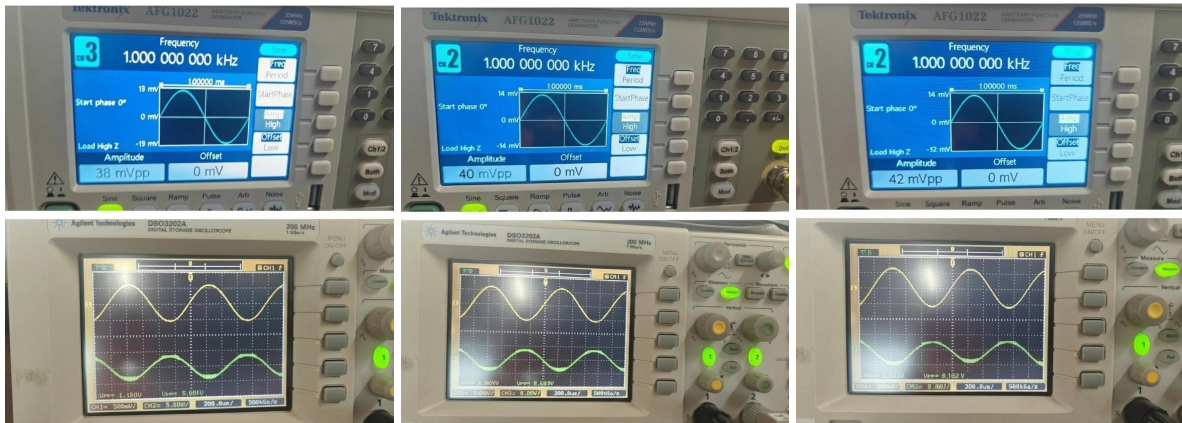
Input V _{pp}	Output V _{pp}	Output V _{rms}	Power (W)
20 mV	3.121	1.103	0.152
21 mV	3.321	1.175	0.172
23 mV	3.561	1.260	0.198
24 mV	3.721	1.316	0.216
25 mV	3.881	1.372	0.235
26 mV	4.481	1.585	0.314
27 mV	4.561	1.613	0.325
28 mV	4.721	1.670	0.348
29 mV	4.900	1.733	0.375
30 mV	4.961	1.755	0.385
35 mV	5.521	1.953	0.477
36 mV	5.781	2.044	0.522
37 mV	5.800	2.051	0.525
38 mV	5.800	2.051	0.525
39 mV	5.921	2.095	0.548
40 mV	6.001	2.123	0.563
41 mV	6.081	2.151	0.578
42 mV	6.161	2.180	0.594
45 mV	6.161	2.180	0.594



(a) 28 mV input

(b) 30 mV input

(c) 35 mV input



(d) 38 mV input

(e) 40 mV input

(f) 42 mV input (clipping)

Figure 6: Output waveforms at different input amplitudes showing mid-level amplification, onset of clipping, and full clipping.

6.4 Total Harmonic Distortion (THD)

The THD was measured at 1 kHz using the oscilloscope FFT function. The amplitudes of the fundamental and the first two harmonics were recorded and are shown in Table 3.

$$THD = \frac{\sqrt{V_2^2 + V_3^2}}{V_1}$$

$$THD = \frac{\sqrt{(0.20)^2 + (0.027)^2}}{1.36} = 0.148 \approx 14.8\%$$

Table 3: Measured harmonic components for THD calculation

Component	Amplitude (V)
Fundamental (1 kHz)	1.36
2nd Harmonic (2 kHz)	0.20
3rd Harmonic (3 kHz)	0.027

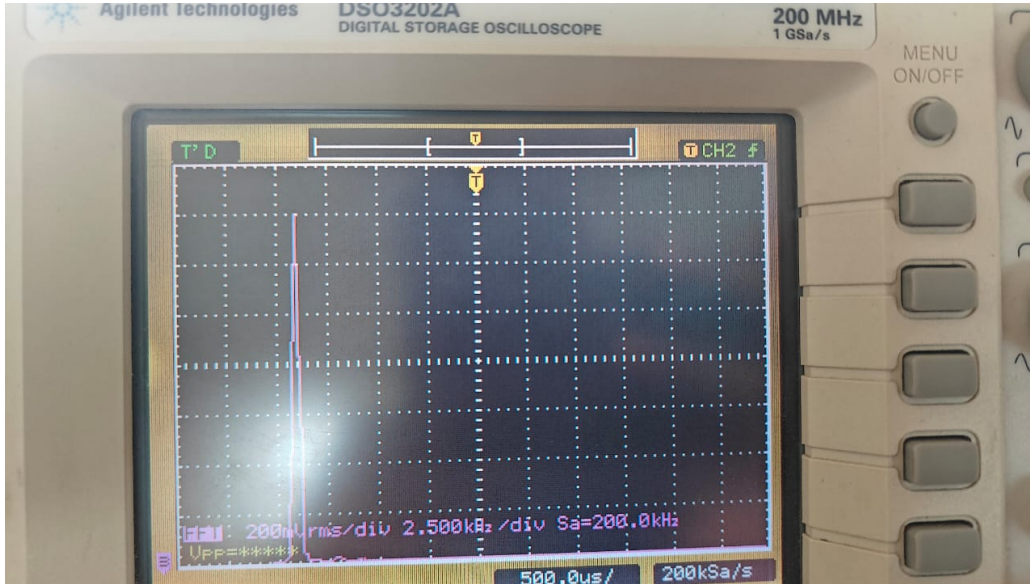


Figure 7: FFT showing the 1 kHz fundamental component.

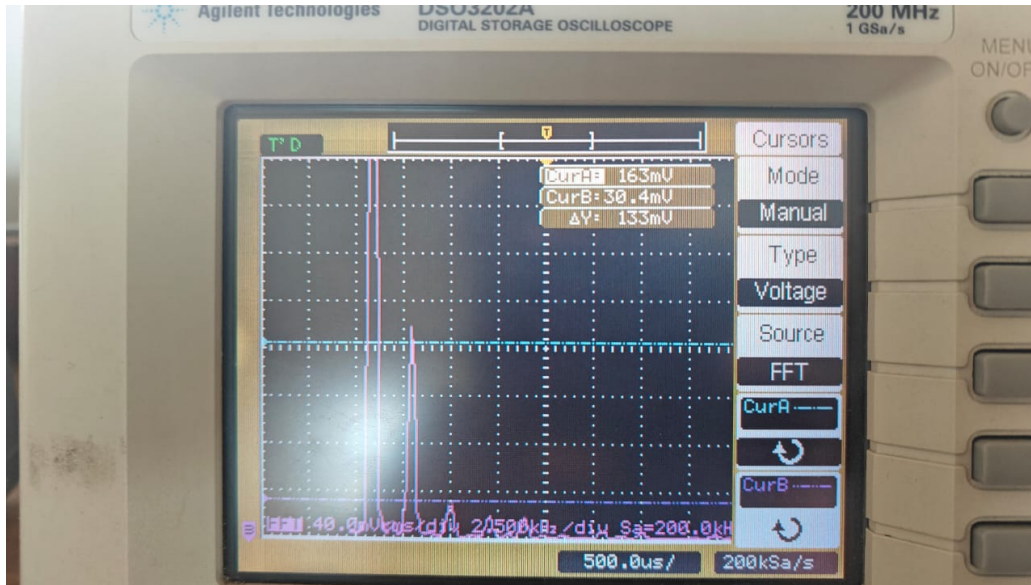


Figure 8: FFT showing the 2 kHz second harmonic.

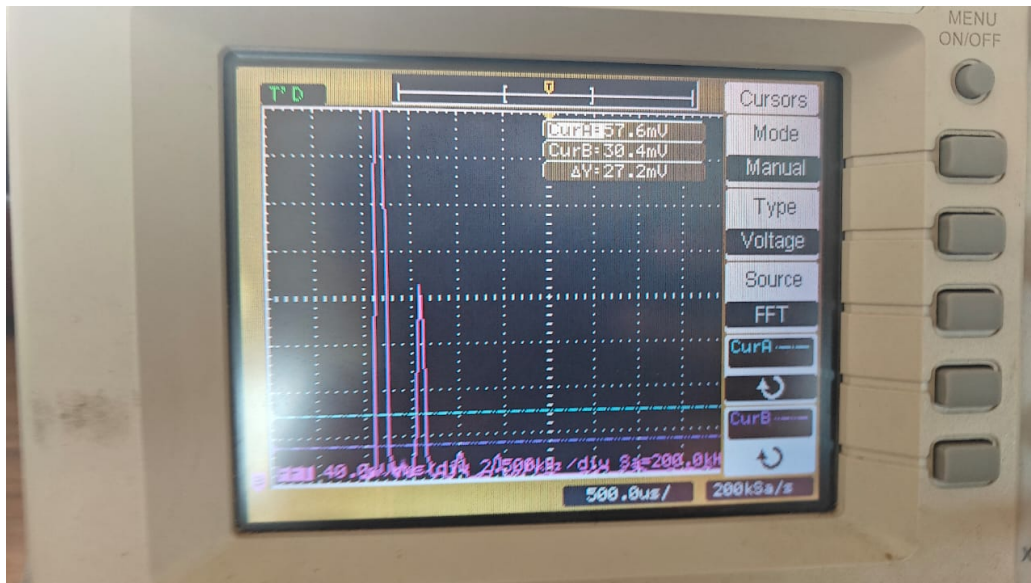


Figure 9: FFT showing the 3 kHz third harmonic.

6.5 Noise

The input of the amplifier was shorted to ground through the coupling capacitor, and the output was observed in AC-coupled mode to measure the inherent noise. Multiple readings were taken to ensure consistency. The measured peak-to-peak noise values were:

$$42 \text{ mV}, \quad 40 \text{ mV}, \quad 32 \text{ mV}$$

The average noise level is:

$$V_{\text{noise,avg}} = 38 \text{ mV (approx.)}$$

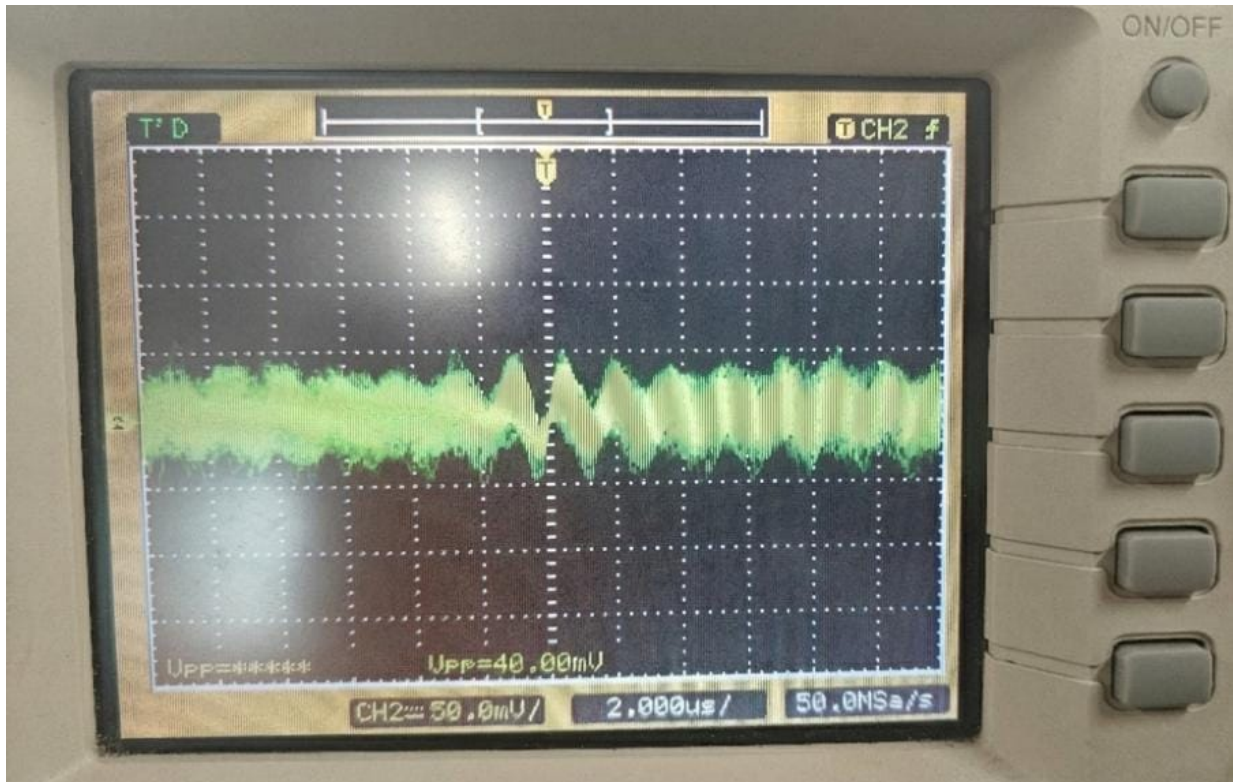


Figure 10: Oscilloscope measurement of output noise with input shorted.

6.6 Power Supply Rejection Ratio (PSRR)

A 16 mV_{pp} ripple was injected into the 12 V supply line using a coupling capacitor. The resulting ripple observed at the amplifier output was 80 mV_{pp} . The output ripple includes both transferred supply ripple and the inherent noise of the amplifier, and therefore no noise subtraction was applied.

$$PSRR = 20 \log_{10} \left(\frac{16}{80} \right) = -14 \text{ dB}$$



Figure 11: Measurement of supply ripple and corresponding output ripple.

7 Results

1. Quiescent Current (I_Q)

The Class AB output stage drew a quiescent current of:

$$I_Q = 98.7 \text{ mA}$$

2. Voltage Gain and Bandwidth

The amplifier provided a midband voltage gain of approximately:

$$A_v \approx 150$$

The -3 dB point occurred around 600 kHz, giving a bandwidth of:

$$BW \approx 20 \text{ Hz to } 600 \text{ kHz}$$

3. Output Swing and Power

The maximum undistorted output swing was:

$$V_{o,\max} \approx 6.0 \text{ V}_{pp}$$

Corresponding to a maximum output power of:

$$P_{\max} \approx 0.56 \text{ W (at } 8 \Omega)$$

4. Total Harmonic Distortion (THD)

Using FFT at 1 kHz, the measured THD was:

$$THD \approx 14.8\%$$

5. Noise

The output noise with input shorted was observed in the range:

$$32 \text{ mV}_{pp} \text{ to } 42 \text{ mV}_{pp}$$

with an approximate average of:

$$V_{\text{noise,avg}} \approx 38 \text{ mV}_{pp}$$

6. Power Supply Rejection Ratio (PSRR)

A supply ripple of 16 mV_{pp} produced an output ripple of 80 mV_{pp} , giving:

$$PSRR = -14 \text{ dB}$$

8 Conclusion

- A microphone pre-amplifier and Class AB power stage were successfully designed and tested.
- The pre-amplifier provided sufficient gain to raise the microphone signal to a usable level.
- The Class AB push–pull stage delivered clear audio output and could drive a low-impedance load.
- The amplifier achieved stable biasing with a quiescent current of approximately 98.7 mA .
- The measured midband gain was around 150, with a bandwidth extending up to about 600 kHz .
- The maximum undistorted output swing was about 6 V_{pp} , corresponding to roughly 0.56 W of output power.
- THD, noise, and PSRR measurements matched the expected performance for a discrete Class AB design.
- Overall, the amplifier operated reliably and met the objectives of signal amplification and characterization.